Project Design Phase
Solution Architecture

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Project Name	Heart Disease Analysis
Maximum Marks	2 Marks

Solution Architecture

Solution architecture is a complex process — with many sub-processes — that bridges the gap between business problems and technology solutions. For the Heart Disease Analysis project, the solution architecture describes how raw patient health data flows from a CSV dataset through a MySQL database into Tableau visualizations and is ultimately published as an interactive dashboard on Tableau Public.

Its goals are to:

- Find the best technology solution to solve the problem of unstructured, scattered heart disease data.
- Describe the structure, characteristics, behavior, and data flow of the Tableau visualization system to project stakeholders.
- Define features, development phases, and solution requirements for the dashboard.
- Provide specifications according to which the solution is defined, managed, and delivered to end users.

Architecture Layers & Data Flow

The Heart Disease Analysis system follows a 5-layer architecture — from raw data ingestion to enduser interaction. Each layer has a specific role in transforming raw CSV data into meaningful, actionable Tableau dashboards.

Layer	Component	Description	Technology	Output
Layer 1	Data Source	Heart_new2.csv — 4,500 patient records, 18 columns including HeartDisease, BMI, Smoking, Age, Gender, Race, Diabetic, and more	Local Filesystem	Raw CSV Input
Layer 2	Data Processing	Data imported and stored in MySQL database. SQL queries used to clean, filter, and transform data for Tableau connection	MySQL Database	Structured Storage
Layer 3	BI & Visualization	Tableau Public connects to MySQL. 10 interactive visualizations created across sheets including bar charts, treemaps, bubble charts, pie charts, and donut charts	Tableau Public	Visual Analytics
Layer 4	Dashboard & Story	All sheets assembled into a Tableau Dashboard and Story with interactive filters for age, gender, race, lifestyle habits, and disease co-occurrence	Tableau Story	Interactive Output
Layer 5	End Users	Published to Tableau Public. Stakeholders — Dr. Sharma, Ramesh, and Anita — access the live dashboard via browser for clinical, policy, and personal health insights	Tableau Cloud	Stakeholder Access

Architecture Flow Description

Step 1 — Data Ingestion

The raw dataset Heart_new2.csv containing 4,500 patient health records across 18 columns is downloaded and validated. Columns include HeartDisease (target), BMI, Smoking, AlcoholDrinking, Stroke, PhysicalHealth, MentalHealth, DiffWalking, Sex, AgeCategory, Race, Diabetic, PhysicalActivity, GenHealth, SleepTime, Asthma, KidneyDisease, and SkinCancer.

Step 2 — Database Storage & SQL Operations

The dataset is imported into a MySQL database using structured SQL operations. Tables are created with appropriate data types for each column. SQL queries are used to filter, aggregate, and prepare data subsets for Tableau connection. Key operations include SELECT with WHERE clauses for filtering by HeartDisease = Yes, GROUP BY for demographic analysis, and JOIN operations for co-occurrence analysis.

Step 3 — Tableau Connection & Visualization

Tableau Public connects directly to the MySQL database using a live or extracted data connection. Ten visualization sheets are created covering Gender vs Heart Disease, Age vs Heart Disease, BMI & Diabetic analysis, Smoking & Alcohol impact, Race-wise distribution, Physical Activity, General Health, Stroke co-occurrence, Diabetic vs Stroke, and Stroke vs Other Diseases.

Step 4 — Dashboard & Story Assembly

All ten visualization sheets are assembled into an interactive Tableau Dashboard and Story. Interactive filters allow stakeholders to drill down by Age Category, Sex, Race, Smoking status, and Diabetic condition. The Tableau Story titled 'Heartbeat of Data' narrates the analysis journey from demographic overview to lifestyle risk factor deep-dives.

Step 5 — Publication & Stakeholder Access

The final workbook is published to Tableau Public as a packaged workbook (.twbx). Dr. Sharma accesses clinical risk dashboards, Ramesh uses demographic and lifestyle analytics for policy work, and Anita views personalized health risk comparisons — all via browser without requiring Tableau Desktop installation.

Architecture Summary

The Heart Disease Analysis solution architecture is designed to be lightweight, publicly accessible, and easily scalable. The five-layer pipeline — CSV → MySQL → Tableau → Dashboard → Tableau Public — ensures a clean separation of concerns between data storage, analysis, and visualization, while remaining simple enough for non-technical stakeholders to interact with the final output.